Pierre Nassivera
pierre.nassivera.1@ulaval.ca
June 2026

Cover letter

Dear Editors of *Biological Conservation*,

We are pleased to submit our manuscript entitled *"Wind energy development alters breeding habitat of the Bicknell's Thrush, a threatened range-restricted boreal songbird"* to *Biological Conservation* for consideration.

The rapid expansion of renewable energy infrastructure is widely recognized as an important component of climate mitigation strategies, yet it also raises significant challenges for biodiversity conservation. In our study, we examine how wind energy development affects breeding habitat for the Bicknell's Thrush (*Catharus bicknelli*), a threatened boreal songbird classified as Vulnerable by the IUCN, and one of North America's most geographically restricted breeding songbirds.

Focusing on a region undergoing large-scale wind energy development since 2013 in Québec (Canada), located at the core of the species' breeding range, **our study provides the first quantitative assessment of how wind energy infrastructure alters breeding habitat for the Bicknell's Thrush**, with impacts disproportionately affecting the highest-priority habitat.

Importantly, **our study also demonstrates that significant opportunities existed within the study area to achieve the same wind energy capacity while avoiding impacts on the most sensitive habitat**, a result that holds even under scenarios favorable to wind energy development. Notably, two of the four wind farm sectors examined account for only a very small share of the total impact (despite having the same theoretical energy production). This contrast indicates that ecological impacts could be avoided and critically depend on spatial siting decisions. By identifying opportunities to avoid impacts before development occurs, our study demonstrates how conservation objectives can be integrated more effectively into infrastructure planning.

Finally, we believe that these results are particularly relevant as wind energy projects and other energy infrastructures continue to expand across Québec and worldwide. Beyond the conservation of the Bicknell's Thrush, **our study provides a practical example of how high-resolution ecological information can be used early in the planning process to identify opportunities to reduce biodiversity impacts** before major infrastructure projects are built, not only for this species but also for other species of conservation concern facing similar land-use conflicts.

We therefore believe that our manuscript will be of interest to the readership of *Biological Conservation* by demonstrating how conservation principles can directly inform real-world

planning decisions and help avoid unnecessary biodiversity loss and its associated consequences.

This manuscript has not been published previously and is not under consideration for publication elsewhere. We declare that we have no competing interests.

Thank you for your consideration.

We look forward to your reply,

Pierre Nassivera,
Corresponding author



Full list of authors and their affiliations

Pierre Nassivera[1], Francis Lessard[1], Hugues Dorion[2], Jérôme Lemaître[3], Yves Aubry[4], Junior A. Tremblay[1,5]

[1]Département des Sciences du bois et de la Forêt, Université Laval, Québec, Canada
[2]Centre d'étude de la Forêt, Département des Sciences fondamentales, Université du Québec à Chicoutimi, Chicoutimi, Canada
[3]Ministère de l'Environnement, de la Lutte contre les changements climatiques, de la Faune et des Parcs, Québec, Canada
[4]Canadian Wildlife Service, Environment and Climate Change Canada, Québec, Canada
[5]Biodiversity Research Division, Environment and Climate Change Canada, Québec, Canada

Environment and Climate Change Canada
Environnement et Changement climatique Canada

Environnement, Lutte contre les changements climatiques, Faunes et Parcs
Québec

cfr
Centre for Forest Research